# INTEGRAL QUASI-COHERENT ALGEBRAS ARE DIRECTED COLIMITS OF FINITE FINITELY PRESENTED ONES

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 2691-2749, following the statement at 2678-2689.

*Editorial note.* Two 'see proof of Lemma' references point to the displayed proof of the sibling lemma; flagged in the notes.

*Allowed standard prerequisites (editorial).*

- Colimit over finite type subalgebras. lemma-algebra-directed-colimit-finite-type (same chapter, cited at ~2692-2695). A = colim A_i over finite type quasi-coherent subalgebras.
- Finite type plus integral equals finite. Algebra, Lemma algebra-lemma-characterize-finite-in-terms-of-integral (cross-chapter, cited at ~2697-2699). Finite type subalgebras of an integral algebra are finite, checked on affine etale covers.
- Colimit of finitely presented modules. lemma-colimit-finitely-presented (same chapter, cited at ~2702-2704 and ~2714-2715). A = colim F_i and J_i = colim E_{ik}.
- Directedness of the kernel system. Cohomology of Spaces, Lemma spaces-cohomology-lemma-finite-presentation-quasi-compact-colimit (cross-chapter, cited at ~2725-2727). Lifts E_{ik} into E_{i'k'} compatibly with J_i -> J_{i'}.
- Displayed proof of the sibling lemma. lemma-finite-algebra-directed-colimit-finite-finitely-presented (same chapter, cited at ~2739-2740 and ~2745-2746). Finiteness and finite presentation of A_{ik} for k large, and the first colimit equality.

*Source reference.* Original reference 02JJ: focused-seeds/geometry/S9-spaces-limits-integral-algebra-directed-colimit-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 082B: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2618–2618. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 07V9: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2492–2492. Navigation pointer only: the counted proof cites

*Date*: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 086Z: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2639–2639. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 07U7: focused-seeds/geometry/S9-spaces-limits-integral-algebra-directed-colimit-finite/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be a quasi-compact and quasi-separated algebraic space over $S$. Let $\mathcal{A}$ be an integral quasi-coherent $\mathcal{O}_X$-algebra. Then*

(1) *$\mathcal{A}$ is the directed colimit of its finite quasi-coherent $\mathcal{O}_X$-subalgebras, and*
(2) *$\mathcal{A}$ is a directed colimit of finite and finitely presented $\mathcal{O}_X$-algebras.*

*Proof.* By Lemma 082B we have $\mathcal{A} = \operatorname{colim} \mathcal{A}_i$ where $\mathcal{A}_i \subset \mathcal{A}$ runs through the quasi-coherent $\mathcal{O}_X$-sub algebras of finite type. Any finite type quasi-coherent $\mathcal{O}_X$-subalgebra of $\mathcal{A}$ is finite (use Algebra, Lemma 02JJ on affine schemes étale over $X$). This proves (1).

To prove (2), write $\mathcal{A} = \operatorname{colim} \mathcal{F}_i$ as a colimit of finitely presented $\mathcal{O}_X$-modules using Lemma 07V9. For each $i$, let $\mathcal{J}_i$ be the kernel of the map

$$\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i) \longrightarrow \mathcal{A}$$

For $i' \geq i$ there is an induced map $\mathcal{J}_i \to \mathcal{J}_{i'}$ and we have $\mathcal{A} = \operatorname{colim} \operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)/\mathcal{J}_i$. Moreover, the quasi-coherent $\mathcal{O}_X$-algebras $\operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)/\mathcal{J}_i$ are finite (see above). Write $\mathcal{J}_i = \operatorname{colim} \mathcal{E}_{ik}$ as a colimit of finitely presented $\mathcal{O}_X$-modules. Given $i' \geq i$ and $k$ there exists a $k'$ such that we have a map $\mathcal{E}_{ik} \to \mathcal{E}_{i'k'}$ making

$$\begin{array}{ccc} \mathcal{J}_i & \longrightarrow & \mathcal{J}_{i'} \\ \uparrow & & \uparrow \\ \mathcal{E}_{ik} & \longrightarrow & \mathcal{E}_{i'k'} \end{array}$$

commute. This follows from Cohomology of Spaces, Lemma 07U7. This induces a map

$$\mathcal{A}_{ik} = \operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_i)/(\mathcal{E}_{ik}) \longrightarrow \operatorname{Sym}^*_{\mathcal{O}_X}(\mathcal{F}_{i'})/(\mathcal{E}_{i'k'}) = \mathcal{A}_{i'k'}$$

where $(\mathcal{E}_{ik})$ denotes the ideal generated by $\mathcal{E}_{ik}$. The quasi-coherent $\mathcal{O}_X$-algebras $\mathcal{A}_{ki}$ are of finite presentation and finite for $k$ large enough (see proof of Lemma 086Z). Finally, we have

$$\operatorname{colim} \mathcal{A}_{ik} = \operatorname{colim} \mathcal{A}_i = \mathcal{A}$$

Namely, the first equality was shown in the proof of Lemma 086Z and the second equality because $\mathcal{A}$ is the colimit of the modules $\mathcal{F}_i$. □